JOMAR L. RANGASAJO (BSChE-1B) PORTFOLIO 5: STRINGS & FUNCTIONS (Chemical Chain Generator)

Introduction: In materials engineering and chemistry, molecules like polymers are created by repeating smaller chemical blocks (monomers) over and over again to form a long chain. This program uses String Arithmetic. It uses the multiplication operator (*) to automatically repeat a chemical compound name a specific number of times, and the addition operator (+) to glue a terminating molecule onto the end of the chain, returning the final synthetic structure back to the user.

In [1]:

```
def build_polymer_chain(monomer, links, terminator):
    full_chain = (monomer * links) + terminator
    return full_chain
user_monomer = input("Enter the repeating chemical unit: ")
repeat_count = int(input("Enter how many units to link together: "))
user_end_piece = input("Enter the terminating molecule for the end: ")
result_chain = build_polymer_chain(user_monomer, repeat_count, user_end_piece)
print("Synthesis Complete")
print("Monomer Base Used:", user_monomer)
print("Total Chain Length:", repeat_count, "units")
print("Generated Structure:", result_chain)
```

```
Synthesis Complete
Monomer Base Used: CH2
Total Chain Length: 4 units
Generated Structure: CH2CH2CH2CH2OH
```

In [ ]: